

$$1) \begin{aligned} a_1 &= 100 \\ a_2 &= 98 \\ a_n &= ? \end{aligned} \quad d = -2$$

$$a_{30} = 100 + (n-1) \cdot d$$

$$a_{30} = 100 - 58$$

$$a_{30} = 42$$

$$s_{30} = \frac{30}{2} (2 \cdot 100 - (29) 2)$$

$$30 (100 - 29)$$

$$30 \cdot 71$$

$$2130$$

$$2) \begin{aligned} a_1 &= 1000000 \\ a_2 &= 10000000 + (10000000 \cdot 0,10) \\ a_n &= 10000000 + (1,1)^{n-1} \\ a_5 &= 10000000 + (1,1)^4 \Rightarrow 1464100 \end{aligned}$$

$$s_5 = (a_1 + a_2 + a_3 + a_4 + a_5) \cdot 12$$

$$s_5 = \left[\frac{1000000 (1 - (1,1)^5)}{1 - (1,1)} \right] \Rightarrow 6105100$$

$$3) a_1 + a_2 + a_3 = 36$$

$$P.A = a_1 + a_1 + d + a_1 + 2d = 36$$

$$a_1^2 + (a_1 + d)^2 = (a_1 + 2d)^2$$

$$\text{De 1) } 3a_1 + 3d = 36$$

$$a_2 = 12 \quad \frac{a_1 + d}{a_2 = 12} = 12 \Rightarrow d = 12 - a_1 \quad | \text{ K}$$

$$\text{Ken 2: } a_1^2 + 12^2 = (a_1 + 2(12 - a_1))^2$$

$$a_1^2 + 12^2 = (a_1 + 24 - 2a_1)^2$$

$$a_1^2 + 12^2 = (24 - a_1)^2$$

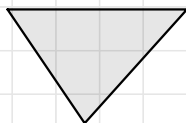
$$\cancel{a_1^2} + 12^2 = 24^2 - 48a_1 + \cancel{a_1^2}$$

$$48a_1 = 24^2 - 12^2$$

$$48a_1 = 432$$

$$a_1 = 9$$

$$a_1 = 9, a_2 = 12 \Rightarrow d = 3 \Rightarrow a_3 = 15$$



$$\left. \begin{array}{l} a_1 = 1 \\ a_2 = 3 \\ a_3 = 4 \\ a_4 = 7 \\ a_5 = 10 \end{array} \right\} d = 3$$
$$a_n = 95$$

$$a_1'' = 4, a_2'' = 7$$

95 Total - 4
91 em P.A

$$S_m = 91$$

$$91 = \frac{m}{2} (2 \cdot 4 + (m-1)3)$$

$$\frac{m}{2} (8 + 3m - 3)$$

$$91 = \frac{m}{2} (5 + 3m)$$

$$182 = 5m + 3m^2$$

$$3m^2 + 5m - 182 = 0$$

$$\frac{-5 \pm \sqrt{25 - 4 \cdot 3 \cdot -182}}{6} \Rightarrow \frac{-5 \pm \sqrt{2209}}{6}$$

$$m_1 = \frac{-52}{6}; m_2 = \frac{42}{6} \Rightarrow 7$$

$$\left. \begin{array}{l} a_1 + a_2 + 7 \end{array} \right) \text{ filas}$$

9 Filas

$$a_1 = 120$$

$$a_n = 120 \left(\frac{6}{7} \right)^{n-1}$$

$$1 = 120 \left(\frac{6}{7} \right)^{n-1}$$

$$\frac{1}{120} = \left(\frac{6}{7} \right)^{n-1} \quad / \quad |_{\ln}$$

$$\ln \left(\frac{1}{120} \right) = (n-1) \cdot \ln \left(\frac{6}{7} \right)$$

$$\frac{\ln \left(\frac{1}{120} \right)}{\ln \left(\frac{6}{7} \right)} = (n-1)$$

$$\frac{\ln \left(\frac{1}{120} \right)}{\ln \left(\frac{6}{7} \right)} + 1 = n$$

$$|3x-2| > x+4$$

$$(3x-2), \quad x \geq \frac{2}{3}$$

$$-(3x-2), \quad x < \frac{2}{3}$$

$$x < \frac{2}{3}; \quad \begin{aligned} -3x+2 &> x+4 \\ 4x+2 &< 0 \\ x &< -\frac{1}{2} \end{aligned}$$

$$x \geq \frac{2}{3}; \quad \begin{aligned} 3x-2 &> x+4 \\ 2x-6 &> 0 \\ x &> 3 \end{aligned}$$

$$S_1 \cap S_2 \Rightarrow]-\infty, -\frac{1}{2}[\cup]3, +\infty[$$

$$3x-2 > x+4 \quad \vee \quad 3x-2 < -x-4$$

$$2x-6 > 0 \quad \vee \quad 4x+2 < 0$$

$$x > 3 \quad \vee \quad x < -\frac{1}{2}$$

$$]-\infty, -\frac{1}{2}[\cup]3, +\infty[$$